Here is the detailed algorithmic extraction and system outline for **Batch 4 (Videos 31 through 40)** of the Trade City Pro technical analysis series.
This batch represents the core operational engine of Arshia Azizpour's strategy: moving from **Volume & Price Action Interaction (VPA)** and the **3-SMA Mechanical Baseline** to the flagship **"Candle Setup" (15m Trigger Engine)**, the **Engulfing Confirmation Model**, and **Direct Order Book Tape Reading**.

Video 31: "آموزش تحلیل تکنیکال پارت 31 : حجم بر اساس پرایس اکشن" (Part 31)

- **Core Concept**: Timeframe inversion of volume data, detecting exhaustion tops/bottoms via Volume-Price divergence, and testing level strength 1-3.
- **Algorithmic Rules & Mechanics**:
- **Lower-Timeframe Volume Precision Rule**: While standard price action patterns are more reliable on higher timeframes (Daily/4H), **volume sensitivity is inversely correlated with timeframe length** 2. The 15-minute chart is optimal for observing institutional order absorption and transaction volume footprints 2.
- **Exhaustion Footprint (Blow-Off / Buying & Selling Climax)** 3, 4:
- When price tests a resistance or support level with massive outlier volume (>3× rolling volume baseline) but fails to penetrate and close past it, institutional absorption is flagged 3.
- *Systematic action*: Do not buy into the volume spike 3. Prepare for an immediate mean-reversion pullback or trend termination 3, 5.
- **Low-Volume Reversal (Lack of Participation)** 5:
- When price crawls into a macro level on contracting, historically low volume, institutional interest has dried up 5. A low-volume test indicates the level cannot be broken without fresh momentum; wait for an opposing displacement trigger 5.
- **The 200-Candle Volume Over-Extension Threshold**:
- Measure the volume envelope over a 200-candle window to establish dynamic ceiling and floor levels for overbought/oversold volume activity 6.

Video 32: "آموزش تحلیل تکنیکال پارت 32 : حجم از نگاهی دیگر بر اساس پرایس اکشن" (Part 32)

- **Core Concept**: Volume Profile (Volume-at-Price) paired with Volume-at-Time boxes to construct multi-dimensional high-liquidity nodes 7-9.
- **Algorithmic Mechanics**:
- **Volume-at-Price Profiling**:
- Run a 200-candle horizontal volume profile to map the heaviest traded value clusters 8.
- These high-volume nodes define natural institutional support and resistance barriers where large orders will be executed 8.
- **Time-and-Price Intersection Box Model**:
- Map horizontal price bands that contain peak volume density 8, 9.
- Map vertical time segments where rolling trading activity expanded beyond standard thresholds 9.
- The overlapping coordinate rectangles form **Liquidity Reaction Boxes** 9.

- **Multi-Timeframe Calibration Workflow**:
- Construct the initial Liquidity Reaction Boxes on the Daily chart 8, 10.
- Step down to the 4-Hour chart and adjust the box boundaries to the most precise volume swing extremes 10.
- Step down to the 15-Minute execution chart: a decisive breakout past the calibrated box boundary signals high momentum, validating a trigger entry 11, 12.

Video 33: "آموزش تحلیل تکنیکال پارت 33 : رابطه حجم با کندل ها بر اساس پرایس اکشن" (Part 33)

- **Core Concept**: Candlestick range vs. tick volume relationships (Effort vs. Result) 13, 14.
- **Algorithmic Metrics & Anomaly Filters**:
- **High Effort, Zero Result (Absorption Anomaly)**:
- *Condition*: A candlestick displays an abnormally large volume bar, but its real body is compressed (small range / doji) 14.
- *Deduction*: Aggressive market orders were absorbed by massive passive limit order walls (Makers) 14. In an uptrend, this signals hidden institutional distribution; in a downtrend, it signals accumulation 14.
- **Low Effort, High Result (Liquidity Vacuum)**:
- *Condition*: A candlestick displays an outsized real body on light, below-average volume 14, 15.
- *Deduction*: The book is empty on the opposing side; price is sliding through a liquidity vacuum. Slippage risk is elevated, and a sharp re-test will follow once resting liquidity returns 15.
- **Volume Displacement Confirmation**:
- Authentic breakouts require expanding volume coupled with an expanding candlestick real body closing near its extreme 14.

Video 34: "آموزش تحلیل تکنیکال پارت 34 : ترید با 3 تا موونیگ اورج بر اساس پرایس اکشن" (Part 34)

- **Core Concept**: The core 3-SMA system (7, 25, 99) as a trend directional governor, dynamic trailing structure, and mean-reversion anchor 16, 17.
- **Parameter Settings**:
- **7 SMA (Simple Moving Average)**: Immediate momentum vector and dynamic candle anchor 17.
- **25 SMA**: Intermediate trend structure and baseline support/resistance 17.
- **99 SMA**: Macro cycle baseline and gravitational anchor ("The Black Hole") 17.
- **Algorithmic Rules**:
- **Alignment Architecture**:
- *Bullish Regime*: Price > 7 SMA > 25 SMA > 99 SMA.
- *Bearish Regime*: Price < 7 SMA < 25 SMA < 99 SMA.
- **The 99 SMA Gravity Filter**:

- When price extends excessively far from the 99 SMA, mean-reversion pressure increases 17, 18. The algorithm forbids initiating new breakout trades in the direction of the extension until price either pulls back to the 99 SMA or consolidates long enough for the 99 SMA to catch up 17.
- **Slope Degradation Governor**:
- When the slope of the 7 SMA transitions from steep to horizontal (slope \\(\approx 0\\)), trend velocity has ceased; lock in profits and halt new entries 18.

Video 35: "آموزش تحلیل تکنیکال پارت 35 : کندل ستاپ ترید ارشیا عزیزپور بر اساس پرایس اکشن و کندل شناسی" (Part 35)

- **Core Concept**: Introduction to the primary proprietary "Candle Setup" on the 15-minute timeframe 19-21.
- **Setup Criteria & Algorithmic Filters**:
- **Contextual Placement**: Must occur strictly in the direction of the dominant higher-timeframe trend 21, 22.
- **Second Wave Constraint**: Must occur during the **second impulse wave** following an initial breakout or structural shift 21, 22. The first leg is allowed to escape; entry occurs on the continuation leg 21, 22.
- **Candle Geometry**:
- Rejection wick: Long wick extending opposite to trade direction (bottom wick for Longs, top wick for Shorts) 22, 23.
- The opposite advance wick must be minimal or absent (smaller than the real body) 22.
- **The 7 SMA Anchor**:
- The 7 SMA line must sit behind the candle, supporting the rejection wick 22.
- **Invalidation**: The 7 SMA line must **never** slice through the real body of the setup candle 22.
- **Execution Logic**:
- Stop-Loss is placed structurally behind the extreme tip of the rejection wick 22, 23.
- Minimum baseline Risk-to-Reward: \\(1:2\\), targeting \\(1:4\\) to \\(1:5\\) 19, 22.

Video 36: "آموزش تحلیل تکنیکال پارت 36 : کندل ستاپ ترید ارشیا عزیزپور بر اساس پرایس اکشن و کندل شناسی پارت 2" (Part 36)

- **Core Concept**: Multi-indicator confluence rules for the 15-minute Candle Setup (RSI Surges, S/R Breaches, and Volume Calibration) 24-26.
- **Quantitative Confluence Checklist**:
- **Structural S/R Break**: The Candle Setup must form directly at, or breach, an established horizontal key level 24, 27.
- **RSI Momentum Influx**:
- For Longs: RSI (14, Close) must cross into **Overbought (>70)** or be on the verge of penetrating 70 during the candle's formation 24, 25.
- For Shorts: RSI must cross into **Oversold (<30)** 24, 26.

- *Logic*: Entering Overbought/Oversold confirms institutional momentum acceleration, driving rapid displacement and keeping trades short in duration 24, 25.
- **Volume Trajectory**:
- Setup candle itself should have moderate, controlled volume (not an anomalous climax exhaustion bar) 26, 27.
- Rolling volume must be expanding in the direction of the break 26.
- **Timeframe Cascading Factor**: When a 15m RSI enters >70 concurrently with the Candle Setup, monitor the 1H and 4H RSIs for follow-through expansion, allowing winning trades to be held for extended runners 25.

Video 37: "آموزش تحلیل تکنیکال پارت 37 : من ۱۰۰ هزار دلار از کندل ستاپ تو ترید درآوردم" (Part 37) "آموزش کندل ستاپ"

- **Core Concept**: Execution mechanics using Stop-Trigger orders, order management rules, and order book confluence 28-30.
- **Execution & Order Management Engine**:
- **Mandatory Order Type**: **Stop-Buy (Market)** or **Stop-Sell (Market)** 29. Market orders and passive limit orders at candle close are forbidden 29.
- **Trigger Coordinates**:
- *Long Trigger*: \\(\text{High}_{\text{Setup Candle}} + \text{Tick Buffer}\\) 29.
- *Short Trigger*: \\(\text{Low}_{\text{Setup Candle}} - \text{Tick Buffer}\\) 29.
- *Stop-Loss*: Placed behind the opposite extreme wick 30.
- **Order Expiration / Cancellation Rule**:
- The pending trigger order remains valid as long as subsequent candles do not close past the invalidation level 31.
- If a subsequent candle closes its real body beyond the tip of the rejection wick (the stop level) prior to triggering, **cancel the order immediately** 31.
- **Macro 99 SMA Filter**:
- Do not take bearish Candle Setups if price remains above the 99 SMA 32.
- Do not take bullish Candle Setups if price remains below the 99 SMA 32.

Video 38: "آموزش تحلیل تکنیکال پارت 38 : پرایس اکشن کندل شناسی کندل اینگلف" (Part 38)

- **Core Concept**: Engulfing Candlestick pattern mechanics without foreign exchange gap assumptions 33-35.
- **Crypto Adaptation & Algorithmic Rules**:
- **Continuous Auction Reality**: Crypto spot and perpetual markets trade 24/7 with zero standard weekend/overnight gaps 34. An engulfing pattern is validated strictly when the second candle's real body completely covers the entire previous candle's body and wicks 34, 35.
- **Contextual Weakness Prerequisite**:
- Never trade an engulfing candle in isolation 36, 37.
- Prior price action must show explicit signs of trend weakness (contracting volume on previous pushes, multiple boundary rejections, or negative divergence) 36, 38.
- **Trigger Execution**:

- *Bearish Engulfing*: Place a Stop-Sell order at \\(\text{Low}*{\text{Engulfing Candle}} - \text{Buffer}\\); *Stop-Loss at* \\(\text{High}*{\text{Engulfing Candle}}\\) 35.
- *Bullish Engulfing*: Place a Stop-Buy order at \\(\text{High}*{\text{Engulfing Candle}} + \text{Buffer}\\); *Stop-Loss at* \\(\text{Low}*{\text{Engulfing Candle}}\\) 38.

Video 39: "آموزش تحلیل تکنیکال پارت 39 : اوردر بوک بر اساس پرایس اکشن در ترید ارزدیجیتال" (Part 39)

- **Core Concept**: Microstructure analysis using the Level 2 Order Book, Depth Charts, and Real-Time Market Taker flows 39-41.
- **Algorithmic Mechanics**:
- **Book Composition**:
- Distinguish between **Passive Orders (Makers)** resting on limit bid/ask ladders and **Aggressive Orders (Takers)** executing instantly across the spread 40-42.
- **Cumulative Depth Aggregation**:
- Monitor aggregate volume depth clusters to calculate how much capital (in USD) is required to displace price by \\(x\%\\) 41, 43.
- **Execution Alignment Rule**:
- When placing a Stop-Buy order above a resistance level, locate the nearest heavy cluster of resting ask limit orders 44.
- Set the trigger point **just above the largest ask wall**, ensuring the position opens in the liquidity vacuum that forms once passive sellers are cleared 44.

Video 40: "آموزش تحلیل تکنیکال پارت 40 : معجزه اُردر بوک آموزش رایگان تحلیل تکنیکال بر اساس پرایس اکشن" (Part 40)

- **Core Concept**: Refining entry precision and stop-loss placement using resting order book walls 45-47.
- **Algorithmic Rules**:
- **Entry Trigger Calibration**:
- If chart price action identifies a resistance level at \\(P_1\\), but order book analysis reveals a massive wall of limit sell orders between \\(P_1\\) and \\(P_2\\), adjust the trigger order to \\(P_2 + \epsilon\\) 45-47.
- Let market buyers absorb the limit wall before entering; entering inside the wall risks rejection before the break 46.
- **Stop-Loss Shielding**:
- When setting a stop-loss for a Long trade, do not place it at an arbitrary technical point 45, 46. Identify the nearest heavy bid cluster (buy wall) and place the stop **just below that wall** 46.
- The resting buy wall serves as structural absorption that institutions must exhaust before the stop can be reached 46.
- **Spoofing / Phantom Liquidity Filter**:
- Real institutional orders remain static in the order book over multiple minutes 48, 49.

- High-frequency algorithmic spoof orders rapidly flash and cancel as price approaches 49. Filter out walls that do not persist across multiple sampling intervals 48, 49.

Progress Tracker

- **Completed Batches**:
- Batch 1: Videos 1–10 (Foundations, Dow Theory, S/R, Capital & Risk Management)
- Batch 2: Videos 11–20 (Market Dominance Ratios, BB on RSI, Risk Expectancy, Profit Ladder)
- Batch 3: Videos 21–30 (Trend Health Analysis, Geometry, Multi-Timeframe Velocity, Volume Dynamics)
- **Batch 4: Videos 31–40 (VPA Modeling, 3-SMA System, 15m Candle Setup Engine, Level 2 Order Books)**
- **Upcoming**:
- **Batch 5 (Videos 41–50)**: Order Book Integration (Parts 41–43), Oscillator Taxonomy, and Methods 1–5 of the 8 Oscillator Framework.

Whenever you are ready, let me know to proceed with **Batch 5 (Videos 41 through 50)**.